

- In the Hubble Space Telescope guide star catalogue, 60% of the objects are binary stars. How large a sample should be chosen to ensure that the probability of the sample containing at least 2 non-binary stars is 99%?

Probability of Non binary star $\Rightarrow IP_N = 0.4$

Let n be the number of non-binary objects in a sample.

We need $IP(n \geq 2) \geq 0.99 \Rightarrow 1 - IP(n < 2) \geq 0.99$

$$IP(n < 2) \leq 0.01$$

↳ We can express $IP(n \leq 1)$ as $IP(n=1) + IP(n=0)$

$$IP(n=0) + IP(n=1) \leq 0.01$$

Now, we can apply the binomial distribution for each case $n=0$ and $n=1$

$$IP(n=0) = \frac{\cancel{N!}}{0! \cdot \cancel{N!}} p^0 (1-p)^N = (1-p)^N$$

$$IP(n=1) = \frac{N!}{1! \cdot (N-1)!} p (1-p)^{N-1} = \frac{NP}{(1-p)} (1-p)^N$$

Where $P = 0.4$ and $1-P = 0.6$

Therefore : $IP(n=0) + IP(n=1) \leq 0.01$

$$(0.6)^N + 0.4 N (0.6)^{N-1} \leq 0.01 \quad \downarrow \text{Hard to solve without approximations.}$$

I made a code for numerically solving it and I got :

↳ check [exercice1.ipynb](#)

$N \geq 14$

- The probability of a certain medical test being positive is 90 %, if a patient has disease D. 1 % of the population has the disease, and the test records a false positive 5 % of the time. If you receive a positive test, what is your probability of having D?

Baye's theorem :
$$IP(D|+) = \frac{IP(+|D) \cdot IP(D)}{IP(+)}$$

Where $IP(+)$: Prob. of receiving a positive test

$IP(D)$: Prob. of having D.

$IP(+|D)$: Prob. of receiving a positive test while having D.

$IP(D|+)$: Prob. of having D while having received a positive test.

$$IP(D) = 0.01$$

$$IP(\tilde{D}) = 0.99$$

$$IP(+|D) = 0.90 \quad \text{True positive}$$

$$IP(+|\tilde{D}) = 0.05 \quad \text{false positive}$$

$$IP(+) = IP(+|D) \cdot P(D) + IP(+|\tilde{D}) P(\tilde{D})$$

$$IP(+) = 0.90 \cdot 0.01 + 0.05 \cdot 0.99$$

$$IP(+) = 0.059$$

Therefore, Bayes theorem becomes :

$$IP(D|+) = \frac{0.90 \cdot 0.01}{0.059}$$

$$IP(D|+) \approx 0.15$$

- In 1929, Edwin Hubble discovered a linear relationship between the distance to a galaxy d and its recession velocity v , given by $v = H_0 d$, where H_0 is known as the Hubble constant. A usual value for $H_0 = 70 \pm 5$ km/s/Mpc. The recession velocity of a galaxy has been measured as $v_m = 100 \pm 5 \times 10^3$ km/s. Determine the posterior probability for the distance to the galaxy in the following cases:

- 1 The value of H_0 is 70 km/s/Mpc.
- 2 The prior for H_0 is:

$$p(H_0|I) = k e^{-\frac{(H_0-70)^2}{2 \times 10^2}}$$

- 3 Assume a uniform prior for H_0 given by:

$$p(H_0|I) = \begin{cases} \frac{1}{90-50} & \text{when } 50 < H_0 < 90 \\ 0 & \text{otherwise} \end{cases}$$

- 4 Assume a Jeffreys prior for H_0 given by:

$$p(H_0|I) = \begin{cases} \frac{1}{H_0 \ln(\frac{90}{50})} & \text{when } 50 < H_0 < 90 \\ 0 & \text{otherwise} \end{cases}$$

1 Case $H_0 = 70 \frac{\text{km/s}}{\text{Mpc}}$

by Bayes theorem, we have : $IP(d|v_m, I) \propto IP(v_m|d, I) IP(H_0|I)$

In this case, we are assuming an uninformative, constant prior

$$IP(H_0|I) = \text{cte} \Rightarrow IP(d|v_m, I) \propto IP(v_m|d, I)$$

The prob. of measuring $v_m = 100 \cdot 10^3$ km/s given the data is modeled by a gaussian likelihood.

$$IP(v_m | d, I) \propto \exp \left\{ - \frac{(v_m - v)^2}{2\sigma_v^2} \right\} \quad v = dH_0 \quad \wedge \quad \sigma_v = 5 \cdot 10^3 \frac{\text{km/s}}{\text{Mpc}}$$

$$IP(v_m | d, I) \propto \exp \left\{ - \frac{(100 \cdot 10^3 - d(70))^2 \frac{\text{km}^2/\text{s}^2}{\text{Mpc}^2} \cdot \frac{\text{Mpc}^2}{\text{km}^2/\text{s}^2}}{2(5 \cdot 10^3)^2} \right\}$$

$$IP(v_m | d, I) \propto \exp \left\{ - \frac{10^2 (10^4/7 - d)^2}{2(5 \cdot 10^3)^2} \right\}$$

$$IP(v_m | d, I) \propto \exp \left\{ - \frac{(1.43 \cdot 10^3 - d)^2}{2(71.4)^2} \right\} \propto IP(d | v_m, I)$$

$$\mu_d = 1.43 \cdot 10^3 \text{ Mpc} \quad \wedge \quad \sigma_d = 71.4 \text{ Mpc}$$



$$d = 1.43 \cdot 10^3 \pm 71.4 \text{ Mpc}$$

* In this simple scenario, the posterior is proportioned to the likelihood so we just have to re-arrange the gaussian likelihood to find the posterior distribution for d given (v_m, H_0) .

2 In this case we have a gaussian prior for $H_0 = k \exp \left\{ - \frac{(H_0 - 70)^2}{2 \cdot 10^2} \right\}$

The Bayes theorem is modified as follows (from last item).

$$IP(d | v_m, I) \propto \int dH_0 \exp \left\{ - \frac{(v_m - v)^2}{2\sigma_v^2} \right\} \exp \left\{ - \frac{(H_0 - \mu_0)^2}{2 \cdot 10^2} \right\}$$

$$IP(d | v_m, I) \propto \int dH_0 \exp \left\{ - \left[\frac{(v_m - dH_0)^2}{2\sigma_v^2} + \frac{(H_0 - \mu_0)^2}{2 \cdot \sigma_H^2} \right] \right\}$$

We can use the model $v = H_0 d$ and squares completion to achieve a gaussian form for the posterior distribution.

$$\begin{aligned} \frac{(v_m - dH_0)^2}{2\sigma_v^2} + \frac{(H_0 - \mu_{H_0})^2}{2\sigma_H^2} &= \frac{v_m^2 + d^2 H_0^2 - 2v_m d H_0}{2\sigma_v^2} + \frac{H_0^2 + \mu_H^2 - 2H_0 \mu_H}{2\sigma_H^2} \\ &= \frac{H_0^2}{2} \underbrace{\left(\frac{d^2}{\sigma_v^2} + \frac{1}{\sigma_H^2} \right)}_{\alpha(d)} - H_0 \underbrace{\left(\frac{v_m d}{\sigma_v^2} + \frac{\mu_H}{\sigma_H^2} \right)}_{\beta(d)} + \frac{1}{2} \underbrace{\left(\frac{v_m^2}{\sigma_v^2} + \frac{\mu_H^2}{\sigma_H^2} \right)}_{\gamma_{cte}} \end{aligned}$$

Where α, β, γ are constants $\alpha \neq \alpha(H_0) \wedge \beta \neq \beta(H_0) \wedge \gamma \neq \gamma(H_0)$

$$\begin{aligned} \frac{(v_m - dH_0)^2}{2\sigma_v^2} + \frac{(H_0 - \mu_{H_0})^2}{2\sigma_H^2} &= \frac{H_0^2}{2} \alpha - H_0 \beta + \frac{\gamma}{2} \\ &= \left(H_0 \sqrt{\alpha/2} - \frac{\beta}{2\sqrt{\alpha/2}} \right)^2 - \frac{\beta^2}{2\alpha} + \frac{\gamma}{2} \end{aligned}$$

$$= \frac{\alpha}{2} \left(H_0 - \frac{\beta}{\alpha} \right)^2 - \underbrace{\frac{\beta^2}{2\alpha} + \frac{\gamma}{2}}_{\text{constants!}}$$

$$\frac{(v_m - dH_0)^2}{2\sigma_v^2} + \frac{(H_0 - \mu_{H_0})^2}{2\sigma_H^2} \propto \frac{\alpha}{2} \left(H_0 - \frac{\beta}{\alpha} \right)^2 \propto \frac{1}{2} \frac{(H_0 - \beta/\alpha)^2}{(\sqrt{\alpha})^2} \rightarrow \text{gaussian shape for } H_0$$

$$\alpha = \left(\frac{d^2}{\sigma_v^2} + \frac{1}{\sigma_H^2} \right) = \frac{d^2 \sigma_H^2 + \sigma_v^2}{\sigma_v^2 \cdot \sigma_H^2} \quad \wedge \quad \beta = \frac{v_m d}{\sigma_v^2} + \frac{\mu_H}{\sigma_H^2} = \frac{v_m d \sigma_H^2 + \mu_H \sigma_v^2}{\sigma_v^2 \sigma_H^2}$$

So, the marginalization over H_0 gives

$$\int_{-\infty}^{\infty} dH_0 \exp \left\{ -\frac{1}{2} \frac{(H_0 - \beta/d)^2}{(\sqrt{1/\alpha})^2} \right\} = \sqrt{1/\alpha} \cdot \sqrt{2\pi}$$

sorry, im using α both
for "proportional to" and as a
function of $\alpha(d)$. I hope there's
no confusion.

$$IP(d | v_m, I) \propto \frac{1}{\sqrt{\alpha}} \cdot \exp \left\{ -\frac{1}{2} \left(\frac{\beta^2}{\alpha} + \gamma \right) \right\}$$

$$-\frac{\beta^2}{\alpha} + \gamma = \frac{(v_m d \sigma_H^2 + \mu_H \sigma_v^2)^2}{(\sigma_v^2 \sigma_H^2)^2} \cdot \frac{\sigma_v^2 \cdot \sigma_H^2}{d^2 \sigma_H^2 + \sigma_v^2} + \gamma = \frac{-1}{\sigma_v^2 \sigma_H^2} \left\{ \frac{(v_m d \sigma_H^2 + \mu_H \sigma_v^2)^2}{(d^2 \sigma_H^2 + \sigma_v^2)} - (v_m^2 \sigma_H^2 + \mu_H^2 \sigma_v^2) \right\}$$

$$-\frac{\beta^2}{\alpha} + \gamma = \frac{-1}{\sigma_v^2 \sigma_H^2 (d^2 \sigma_H^2 + \sigma_v^2)} \left\{ \cancel{v_m^2 d^2 \sigma_H^4} + \cancel{\mu_H^2 \sigma_v^4} + 2v_m d \mu_H^2 \sigma_H^2 \sigma_v^2 - \cancel{v_m^2 d^2 \sigma_H^4} - v_m^2 \sigma_H^2 \sigma_v^2 - \mu_H^2 \sigma_v^2 d^2 \sigma_H^2 - \cancel{\mu_H^2 \sigma_v^4} \right\}$$

$$-\frac{\beta^2}{\alpha} + \gamma \propto \frac{-1}{\cancel{\sigma_v^2 \sigma_H^2} (d^2 \sigma_H^2 + \sigma_v^2)} \left\{ \cancel{\sigma_v^2 \sigma_H^2} \left[2v_m d \mu_H^2 - v_m^2 - \mu_H^2 d^2 \right] \right\}$$

$$-\frac{\beta^2}{\alpha} + \gamma \propto \frac{-1}{\cancel{\sigma_v^2 \sigma_H^2} (d^2 \sigma_H^2 + \sigma_v^2)} \left\{ \cancel{\sigma_v^2 \sigma_H^2} \left[2v_m d \mu_H^2 - v_m^2 - \mu_H^2 d^2 \right] \right\}$$

$$-\frac{\beta^2}{\alpha} + \gamma \propto \frac{1}{2} \frac{(v_m - \mu_H d)^2}{(d^2 \sigma_H^2 + \sigma_v^2)}$$

↳ $IP(d | v_m, I) \propto \frac{1}{\sqrt{\alpha}} \cdot \exp \left\{ -\frac{1}{2} \left(\frac{\beta^2}{\alpha} + \gamma \right) \right\}$

$$IP(d | v_m, I) \propto \frac{\overbrace{\delta_H \delta_v}^{\text{constant}}}{\sqrt{d^2 \delta_H^2 + \delta_v^2}} \exp \left\{ -\frac{1}{2} \frac{(v_m - 70d)^2}{(d^2 \delta_H^2 + \delta_v^2)} \right\}$$

$$IP(d | v_m, I) \propto \frac{1}{\sqrt{d^2 \delta_H^2 + \delta_v^2}} \exp \left\{ -\frac{1}{2} \frac{(v_m - \mu_H d)^2}{(d^2 \delta_H^2 + \delta_v^2)} \right\}$$

Where $\mu_H = 70 \frac{\text{km/s}}{\text{Mpc}}$ $\wedge$ $\delta_H = 10 \frac{\text{km/s}}{\text{Mpc}}$ $\wedge$ $\delta_v = 5 \cdot 10^3 \text{ km/s}$

③ Now we have a "box" prior: $IP(H_0 | I) = \begin{cases} 1/40 & \text{if } 50 < H_0 < 90 \\ 0 & \text{otherwise} \end{cases}$

Using the result from last item, we have that the posterior distribution is:

$$IP(d | v_m, I) \propto \int_{50}^{90} dH_0 \frac{1}{40} \exp \left\{ -\frac{(v_m - dH_0)^2}{2\delta_v^2} \right\}$$

$$IP(d | v_m, I) \propto \int_{50}^{90} \exp \left\{ -\frac{(H_0 - v_m/d)^2}{2\delta_v^2/d^2} \right\} \leftarrow \text{Incomplete Gaussian integral}$$

Using: $\int_a^b dx e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \sqrt{2\pi}\sigma \left[\text{erf}\left(\frac{b-\mu}{\sqrt{2}\sigma}\right) - \text{erf}\left(\frac{a-\mu}{\sqrt{2}\sigma}\right) \right]$

$\hookrightarrow IP(d | v_m, I) \propto \overbrace{\sqrt{2\pi}}^{\text{constant}} \frac{\delta_v}{d} \left[\text{erf}\left(\frac{90 - v_m/d}{\sqrt{2} \delta_v/d}\right) - \text{erf}\left(\frac{50 - v_m/d}{\sqrt{2} \delta_v/d}\right) \right]$

$$IP(d | v_m, I) \propto \frac{1}{d} \left[\operatorname{erf} \left(\frac{90d - v_m}{\sqrt{2} \delta v} \right) - \operatorname{erf} \left(\frac{50d - v_m}{\sqrt{2} \delta v} \right) \right]$$

↳ Where $v_m = 100 \cdot 10^3 \text{ km/s}$ & $\delta v = 5 \cdot 10^3 \text{ km/s}$

④ finally, we have a Jeffreys prior: $IP(H_0 | I) = \begin{cases} 1/H_0 \ln(90/50) & 50 < H_0 < 90 \\ 0 & \text{other} \end{cases}$

from previous results, the posterior distribution will be (ignoring constants as per usual).

$$IP(d | v_m, I) \propto \int_{50}^{90} dH_0 H_0^{-1} \exp \left\{ - \frac{(H_0 - v_m/d)^2}{2\delta v^2/d^2} \right\}$$

This integral can be expressed in terms of the complex (Faddeeva) error function $w(z)$.

↳ Considering: $\int_{-\infty}^{\infty} \frac{e^{-x^2}}{z-x} dx = i\pi w(z)$ where $w(z) = e^{-z^2} \operatorname{erfc}(-iz)$

$$IP(d | v_m, I) \propto \int_{50}^{90} dH_0 H_0^{-1} \exp \left\{ - \frac{(H_0 - v_m/d)^2}{2\delta v^2/d^2} \right\} \quad \text{Let } \frac{(H_0 - v_m/d)}{2\delta v^2/d} = x$$

$$\frac{1}{2\delta v^2/d} dH_0 = dx$$

$$IP(d | v_m, I) \propto \int_{x_a}^{x_b} \frac{1}{(2\delta v^2/d + v_m/d)^{-1}} e^{-x^2} (2\delta v^2/d) dx$$

↳ $(2\delta v^2/d)^{-1} (x + v_m/2\delta v^2)^{-1}$

$$P(d|v_m, I) \propto \frac{d}{d} \int_{x_a}^{x_b} \frac{e^{-x^2}}{(x + \underbrace{v_m/2b_v^2}_{-z})} dx$$

$$P(d|v_m, I) \propto \int_{x_b}^{x_a} \frac{e^{-x^2}}{(z-x)} dx$$

incomplete faddeeva integral
(Don't know if this has an analytical closed form...)